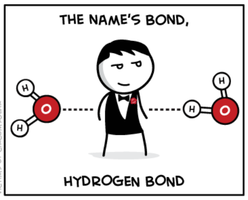
Name ________________________________Period ______Test Day________________



Unit 8 Covalent Bonding Essential Knowledge and Guided Notes

Textbook: CK Chemistry Chapters 7, 8 & 9
https://flexbooks.ck12.org/cbook/ck-12-chemistry-flexbook-2.0/

Helpful Videos and Practice: Posted on CANVAS

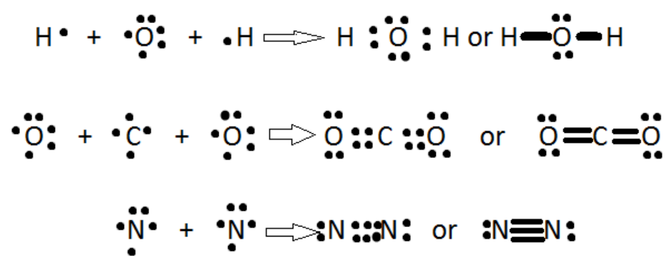
Essential Knowledge:

1. Molecular covalent compounds differ in properties from ionic compounds. While ionic compounds are always solid, molecular compounds can be solids, liquids, or gases.

2. Chemical formulas are used to represent compounds. Subscripts represent the relative number of each type of atom in a molecule or formula unit. The International Union of Pure and Applied Chemistry (IUPAC) system is used for naming compounds.

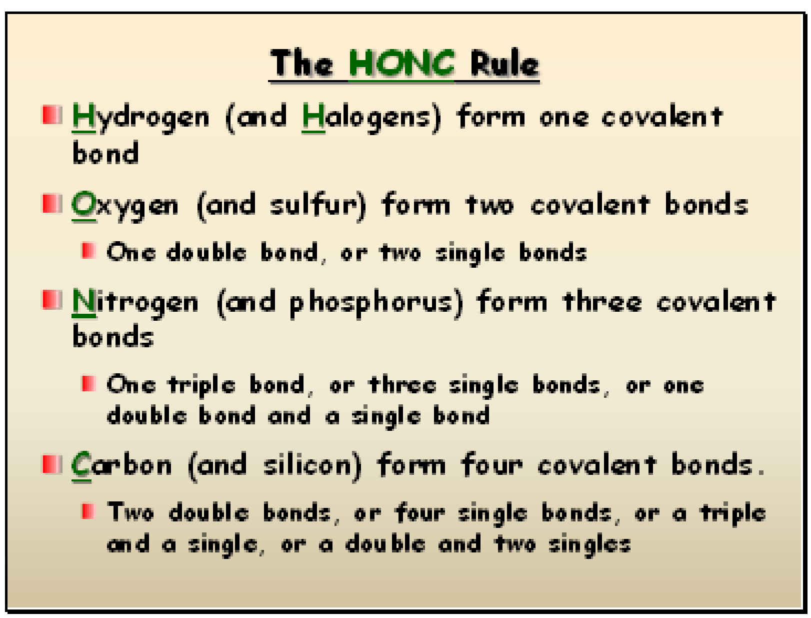
3. Rules for naming (diatomic) binary covalent compounds:
   a. Use a prefix (mono, di, tri, tetra...) up to 10. Exception "mono" on the first element. The least electronegative element first.
   b. The second element ends in –ide
   c. There are a few names that don't follow the rules**. Memorize and write the 3 chemical formulas for these common substances: ammonia (NH_3), water (H_2O), and methane (CH_4)**

4. Bonds form between atoms to achieve stability. Covalent bonds involve the sharing of electrons between atoms. Covalent bonds can be classified as single, double, or triple bonds depending on the number of pairs of electrons shared.
   a. In a single covalent bond, atoms share 1 pair of electrons
   b. In a double covalent bond, atoms share 2 pairs of electrons
   c. In a triple covalent bond, atoms share 3 pairs of electrons

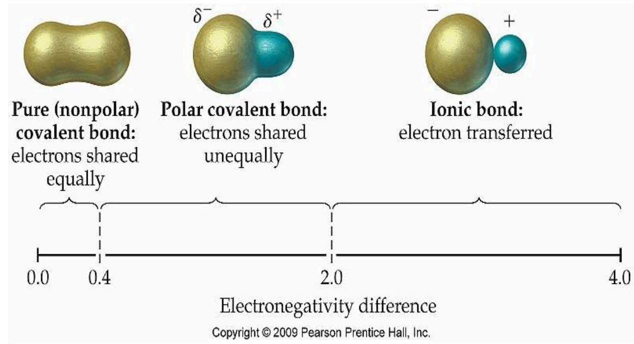


All these diagrams show the formation of covalent bonds as electrons from elements are shared to form single, double, and triple bonds

5. You should be able to explain and do the following:
   a. Explain the meaning of HONC.
   b. Draw Lewis dot diagrams to represent valence electrons in elements and draw Lewis Structural Formulas to show covalent bonding
   c. Lewis dot diagrams are used to represent valence electrons in an element. Structural formulas show the arrangements of atoms and bonds in a molecule and are represented by Lewis dot structures.

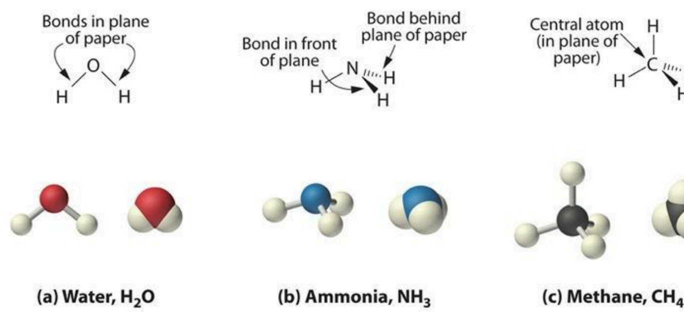


6. The electronegativity of the component atoms in a compound affects the charge distribution in a polar bond.
   a. Non-polar bonds form between elements with similar electronegativities. In a **nonpolar covalent bond,** the electrons are shared equally, and there is no unequal charge distribution.
   b. Polar molecules result when electrons are distributed unequally. In a **polar covalent bond,** electrons are more attracted to the atom with greater electronegativity, resulting in a partial negative charge on that atom. The atom with the smaller electronegativity value acquires a partial positive charge. The general guidelines for telling whether a bond is polar or not:



7. **VSEPR stands for Valence Shell Electron Pair Repulsion**. In covalent compounds, electron pairs arrange themselves in a way to keep them as far apart as possible. You should be able to predict, draw, and name molecular shapes (bent, linear, trigonal planar, tetrahedral, and trigonal pyramidal) and polarities.

8. For molecules with double and triple bonds, you must count the number of bonds to determine the geometry of the bond angles: If carbon is bonded to 2 things it is linear. If carbon is bonded to 3 things it is trigonal planar. If carbon is bonded to four things it is tetrahedral.



Linear "Trigonal" Pyramidal Tetrahedral

9. **The <u>forces of attraction between neighboring molecules</u> are called intermolecular forces or attractions.** They occur between (not within) covalent molecules. Intermolecular attractions are much weaker than ionic and covalent bonds but are responsible for covalent compounds forming liquids and gases.

10. Forces of attraction determine the phase at room temperature. According to the kinetic-molecular theory, the phase of a substance at room temperature depends on the strength of the attraction between its particles. **Molecules with stronger intermolecular forces will have higher boiling points.**

11. There are three main types of intermolecular forces: **<u>Dispersion</u>** forces are the weakest and occur between all covalent molecules. They are caused by temporary dipoles. **<u>Dipole Forces</u>** are stronger and occur between polar molecules. **<u>Hydrogen "bonding"</u>** is a super strong form of dipole forces that occur between molecules that have very large dipoles where hydrogen is directly bonded to the very electronegative elements of F, O, and N.

12. In **network solids**, conventional chemical bonds hold the chemical subunits together. The bonding between chemical subunits, however, is identical to that within the subunits, resulting in a continuous network of chemical bonds. Two common examples of network solids are diamond (a form of pure carbon) and quartz (silicon dioxide)

U8 Lesson #1 - Naming Covalent Compounds - READ AND COMPLETE

Model 1 – Binary Molecular Covalent Compounds

Molecular Formula	Number of Atoms of First Element	Number of Atoms of Second Element	Name of Compound
ClF			Chlorine monofluoride
ClF_5	1	5	Chlorine pentafluoride
CO			Carbon monoxide
CO_2			Carbon dioxide
Cl_2O			Dichlorine monoxide
PCl_5			Phosphorus pentachloride
N_2O_5			Dinitrogen pentoxide

1. Fill in the table to indicate the number of atoms of each type in the molecular formula.
2. Examine the molecular formulas given in Model 1 for various molecular compounds.
   a. How many different *elements* are present in each compound shown? ____
   b. Do the compounds combine metals with metals, metals with nonmetals, or nonmetals with nonmetals? ________________________________
   c. Bonding between nonmetals requires the sharing of electrons. This is known as covalent bonding. Based on your answer to b, what type of bonding must be involved in molecular compounds? ______________________

3. Find all of the compounds in Model 1 that have chlorine and fluorine in them. Explain why the name "chlorine fluoride" is not sufficient to identify a specific compound.

Model 2 – Prefixes and Suffixes

Molecular Formula	Name of Compound
BCl_3	Boron trichloride
SF_6	Sulfur hexafluoride
IF_7	Iodine heptafluoride
NI_3	Nitrogen triiodide
N_2O_4	Dinitrogen tetroxide
Cl_2O	Dichlorine monoxide
P_4O_{10}	Tetraphosphorus decoxide
B_5H_9	Pentaboron nonahydride
Br_3O_8	Tribromine octoxide
ClF	Chlorine monofluoride

Prefix	Numerical Value
mono-	
di-	
tri-	
tetra-	
penta-	
hexa-	
hepta-	
octa-	
nona-	
deca-	

4. Examine the prefixes in Model 2. Fill in the numerical value that corresponds to each prefix.

5. Carefully examine the names of the compounds in Model 2. When is a prefix NOT used in front of the name of the first element? _______________

6. What suffix (ending) do all the compound names in Model 2 have in common? ___________

7. **Consider the compound NO**. Which element, nitrogen or oxygen, would require a prefix in the molecule name? _________ Name the molecule NO _____________________________

8. Find two compounds in Model 2 that contain a subscript of "4" in their molecular formula.
   a) List the formulas and names for the two compounds.

Formula	Name

   b) What is different about the spelling of the prefix meaning "four" in these two names?

9. Find two compounds in Model 2 that contain the prefix "mono-" in their names.
   a) List the formulas and names for the two compounds.

Formula	Name

   b) What is different about the spelling of the prefix meaning "one" in these two names?

10. Identify one remaining name of a compound in Model 2 where the prefixes do not exactly match the spelling shown in the prefix table: ____________________________

11. Use your answers to Questions 8–10 what vowels do you never double or have next to another vowel? ________ and _________. What vowel can you double? ______

12. All of the compounds listed in Model 2 are binary molecular covalent compounds. Compounds such as CH_3OH or PF_2Cl_3 are not binary, but they are also covalent and molecular. What makes a "binary molecular compound" different from just a molecular covalent compound?

13. NaCl or $CaCl_2$ are not binary molecular covalent, but are binary ionic compounds. What makes a "binary molecular covalent compound" different from just a binary ionic compound?

Model 3 – Ionic vs. Binary Molecular Nomenclature

This activity focused on molecular (covalent) compounds, while earlier activities addressed ionic compounds. Notice that the formulas for both types of compounds can look very similar, even though their names are quite different:

Chemical Formula	Type of Compound/Bonding	Compound Name
MgF_2	Ionic	Magnesium fluoride
CuF_2	Ionic	Copper(II) fluoride
SF_2	Molecular (covalent)	Sulfur difluoride
NaBr	Ionic	Sodium bromide
AuBr	Ionic	Gold(I) bromide
IBr	Molecular (covalent)	Iodine monobromide

14. Identify one difference between the names or formulas for ionic compounds versus those for binary molecular compounds.

15. Identify one similarity between the names or formulas for ionic compounds versus those for binary molecular compounds.

Upon completion take the exit ticket in CANVAS. You have 3 tries.

U8 Lesson #2: Covalent Molecules

Characteristics of Covalent Compounds:

1. Covalent compounds are composed of all ______________________________.
2. Covalent compounds can exist as ______________, ________________, _____________
3. Most Covalent Compounds are ____________with_______melting points.
4. A few covalent compounds are _______________ _______________.
5. The smallest unit of ________ covalent compounds is a ______________________
6. Covalent compounds are _______________________________.
7. When dissolved in water they will not______________________. There are no moving charged particles like ionic compounds in water.

Lewis Dot Diagrams to Structural Formulas:

8. For covalent compounds to form, non-metals share electrons, so each atom "feels" it has a full outer energy level. In this way they obey the _________ _____________. When non-metals share a pair of electrons they are said to be "bonded."

9. You will build and draw the Lewis Diagrams sometimes called Lewis Structures for some simple covalent compounds. Elements can share 1 pair (______ electrons), 2 pairs (______ electrons) or 3 pairs (_____ electrons) to feel they have 8 electrons or in the case of Hydrogen 2 electrons. However, elements usually only bond to each other with their unpaired electrons.

10. THE HONC "RULE"

Element	**# Valence Electrons**	**Lewis Diagram**	**# of bonds (How many unpaired electrons in the Lewis Diagram?)**
Hydrogen			
Halogens (F, Cl, Br, I)			
Oxygen (S, Se)			
Nitrogen (P)			
Carbon (Si)			

11. Drawing Lewis Structural Formulas of Covalent Compounds

1. Determine the number of atoms of each element in the molecule.
2. Determine the number of valence electrons for each atom.
3. Add together to find the total number of valence electrons.
4. Arrange the atoms to form a skeleton structure for the molecule. The atom that wants the most bonds (HONC rule) is at the center.
5. Join the atoms with bonding pairs so to fulfill the HONC rule.
6. Add unshared pairs of electrons so that each nonmetal (except hydrogen which is stable with 2 electrons) is surrounded by eight electrons to FULFILL THE OCTET RULE
7. Count the electrons in the structure to be sure that the number of valence electrons used equals the number available. If not, you must consider making double or triple bonds until this requirement is fulfilled, as well as HONC and OCTET.
8. To draw the structural formula, replace bonding pairs of electrons with lines representing bonds:
    a. ::: or ≡ is three shared pairs, a triple covalent bond
    b. :: or = is two shared pairs, a double covalent bond
    c. : or − is one shared pair, a single covalent bond
9. Double Check your Structural Formula.

Example #1: NF_3
\# of Valence Electrons _____

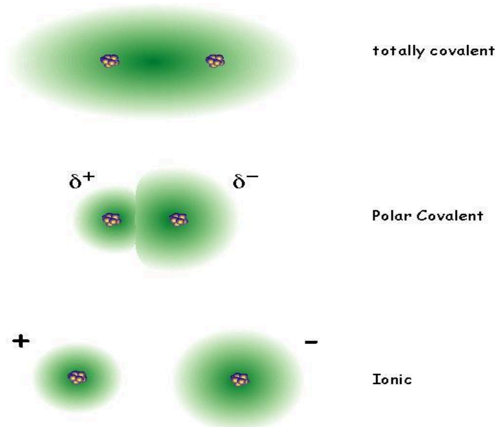
Example #2: (Triple Bonds) HCN
\# of Valence Electrons _____

Example #3: (Double Bonds) CO_2
# of Valence Electrons ______

Example #4 (Exception to Octet) BH_3
# of Valence Electrons ______

U8 Lesson#3: Molecular Geometry

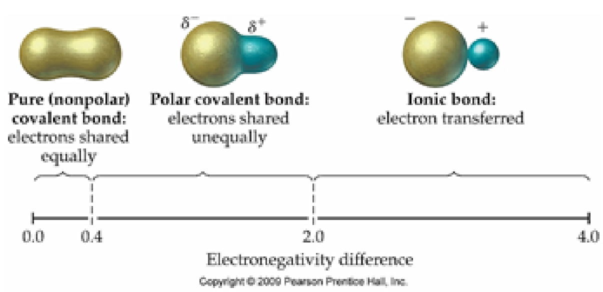
1. Levels of Sharing Electrons:

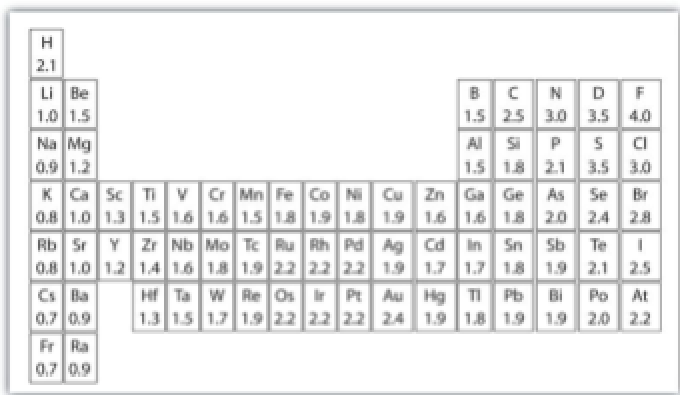


In a ______________________ **covalent bond,** the electrons are shared equally, and there is no unequal distribution of charge

In a ____________________**covalent bond,** electrons are more attracted to the atom with the greater electronegativity, resulting in a partial negative charge on that atom. The atom with the smaller electronegativity value acquires a partial positive charge.

In an _________ **bond,** the electrons are transferred.





H 2.1																
Li 1.0	Be 1.5											B 1.5	C 2.5	N 3.0	D 3.5	F 4.0
Na 0.9	Mg 1.2											Al 1.5	Si 1.8	P 2.1	S 3.5	Cl 3.0
K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.9	Ni 1.8	Cu 1.9	Zn 1.6	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8
Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5
Cs 0.7	Ba 0.9		Hf 1.3	Ta 1.5	W 1.7	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9	Tl 1.8	Pb 1.9	Bi 1.9	Po 2.0	At 2.2
Fr 0.7	Ra 0.9															

<u>Important Note:</u>
Diatomics elements (H_2 O_2 etc.) are always _____, so are always non-polar.
The _______ bond is also considered non-polar   2.5 - 2.1 = 0.4

2. The shapes of molecules may be predicted by using **the VSEPR rule, which states that electron pairs around a central atom will position themselves to allow for the maximum amount of space between them.** (Valence Shell Electron Pair Repulsion)

3. **Table 1: Shapes and Polarities of Covalent Molecules**

# of Electron Groups around the Central Atom	Geometry Name Or Shape and Bond Angle	Structural Formula of Central Atom	Molecular Polarity (Polar or Non-polar)
No central atom. (Diatomics)	**Linear** (no central atom)	X _ X :N ≡ N : :O=O: ·· ··	**<u>Nonpolar</u>** if atoms are the same (O_2, N_2, Br_2) **<u>Polar</u>** if atoms are different (HCl)
2 shared, 0 unshared	**Linear** Double and triple bonds only count as 1 shared pair for geometry purposes Bond angle is 180°	= C = Or ≡ C –	**<u>Nonpolar</u>** if bonded atoms are the same (CO_2) **<u>Polar</u>** if bonded atoms are different (HCN)
3 shared, 0 unshared	**Trigonal planar** Bond angles are 120°	| _ B _ or = C _ |	**<u>Nonpolar</u>** if bonded atoms are the same (BF_3) **<u>Polar</u>** if bonded atoms are different (HCOH)
4 shared, 0 unshared	**Tetrahedral** Since the molecule is 3-dimensional, the bond angles are 109°, not 90°.	| _ C _ |	**<u>Nonpolar</u>** if bonded atoms are the same (CH_4) **<u>Polar</u>** if bonded atoms are different (CH_3F)
2 shared, 2 unshared	**Bent** The bond angle is 105°	| **:O:** |	This shape is not symmetrical so it is **<u>always</u>** POLAR (H_2O or H_2S)
3 shared, 1 unshared	**(Trigonal) Pyramidal** The bond angle is 107°	| _ N**:** |	This shape is not symmetrical so it is **<u>always</u>** POLAR (NH_3 or PH_3)

a. What are the 5 different shapes or geometries that can exist around a central atom? _______________,________________,_______________,_____________ and ________________
b. Which 2 shapes are always polar? ________________ and _______________

c. Which 3 shapes can be polar or non polar based on the terminal atoms bonded to the central atom? ____________, _______________ and ___________________________

<u>U8 Lesson#4: Electronegativity and Polarity (Watch the embedded video)</u>

1. What does a single line stand for in a covalent bond? ___________________

2. What is a covalent bond like? _________________

3. What is the measure of the elements' strength called? _________________

4. What is the definition of electronegativity? _______________________________________

5. What is the most electronegative element? ______

6. What is ΔE for the molecule HCl? ____________

7. Fill in the chart

ΔE (difference in electronegativity values of the two atoms in the bond)	Type of bond (nonpolar covalent, polar covalent or ionic)

8. Is HCl a polar covalent or a non polar covalent bond? _________________

9. Draw the dipole moment for HCl below. Be sure to label as in the video:

H – Cl

10. Does a polar bond necessarily mean the whole molecule will be polar? ____

11. Copy the dipole diagram for CF_4

12. Is the C-F bond polar? ______

13. The CF_4 molecule is non-polar even though the C-F Bond is polar. Why is this according to the video?

<u>U8 Lesson #5 : Intermolecular Forces.</u>

1. The <u>forces of attraction between neighboring molecules</u> are called __________________
2. **These attractions are much ____________________ than either an ionic or a covalent bond.**
3. They occur ____________________ (not within) covalent molecules.
4. Intermolecular Forces are attractions that hold polar and non-polar covalent molecules together so that liquids and solids can form. Forces of Attraction determine the ___________________ at room temperature.
5. According to the kinetic-molecular theory, the phase of a substance at room temperature depends on the strength of the attractions between its particles.
6. **Molecules with stronger intermolecular forces will have higher boiling points.**

- ________________ Intermolecular forces are found in the **gas** phase of matter at STP.
- ______________ Intermolecular forces are found in the **liquid** phase of matter at STP.
- _______________Intermolecular forces are found in the **solid** phase of matter at STP.

7. Three types of intermolecular forces are ____,
   ____________________________________ and _______________________________________

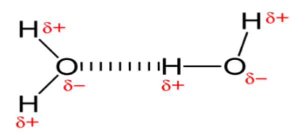
8. Types of Dipoles:

forms when electrons momentarily shift to one side of an atom at random	occur when a neighboring molecule moves past one with a temporary dipole	is the result of a polar bond in the molecular structure

9. What are __________ forces?
   a. Weakest Intermolecular Force (SOL calls them Van der Waals forces)
   b. Caused by temporary dipoles
   c. Present in both nonpolar and polar molecules
   d. The larger the molecule the greater the force and the higher the boiling point

10. What are ________- _________ Forces?
    a. Stronger than Dispersion forces
    b. Form due to Permanent dipoles in Polar Molecules only
    c. Opposite charged ends of the molecules attract

11. What are Hydrogen "Bonds": (why is "bonds" in quotation marks)?
    a. Not really a bond just a very super strong attraction!
    b. Occur only in molecules that have hydrogen bonded to F, O, or N
    c. The strongest intermolecular force
    d. Hydrogen bonds are FON!



Circle the "hydrogen bond" in the picture above

12. What type of intermolecular forces act on non-polar covalent molecules and is the weakest? ________________________________

13. What type of intermolecular forces act between polar molecules that do not contain hydrogen? _________________________ and _______________________________

14. What type of intermolecular forces act between polar molecules that contain Hydrogen bonded to F, O and N? ____________________________